

PLANUM

Smart QR + NFC Stickers — Product Concept v2

One SKU • Three Sizes (25 / 30 / 40 mm) • QR + NFC Combined • Weatherproof Indoor & Outdoor • Two-URL Programming for Auto-Login

This document describes the revised Planum sticker product for the manufacturer enquiry to Shenzhen SPV Field IoT Co., Ltd. It supersedes the v1 concept dated 25 April 2026.

Key changes vs. v1:

- ONE single product variant (SKU) — every sticker combines QR + NFC; no separate variants
- THREE diameters within ONE end-customer pack (25 / 30 / 40 mm) for indoor, herbs and outdoor use
- Hard weather-resistance requirement — outdoor / garden usage is now in scope from day one
- Two-URL programming — QR and NFC each carry a different URL per sticker (registration vs. auto-login)
- Material preference shifted from PP-Vinyl to PET (better outdoor durability, lower environmental footprint)

1 • Product Concept

Planum stickers are the physical bridge between a real plant and its digital care profile. Each sticker carries a unique printed QR code AND a unique pre-programmed NFC tag. The customer attaches the sticker to the plant pot — from that moment, the plant is part of their digital garden in our web app at app.planum.de.

Functional — QR scans must be reliable for at least 3 years across watering, sunlight, soil contact, indoor heating and outdoor weather (rain, frost, summer heat, UV).

Aesthetic — On a beautiful pot the sticker should not feel like a label, but like a discreet brand mark.

Recognisable — Planum brand language: calm green tones, botanical motif, clean typography.

Universal — One product line covers houseplants, herbs, garden plants, balcony planters and outdoor pots.

2 • One SKU, Three Sizes

All stickers share identical material, finish, NFC chip and durability. Only the diameter differs:

Size	Use case
ø 25 mm	Small pots, herbs, propagation cups, seedling trays
ø 30 mm	Standard houseplants — default size, most common
ø 40 mm	Garden plants, outdoor planters, large terracotta pots, raised beds

The end-customer pack contains all three sizes in a fixed assortment (8 × 25 + 8 × 30 + 4 × 40 = 20 stickers per pack). One pack covers everything from a windowsill basil to a balcony lemon tree.

3 • Two-URL Programming — QR vs. NFC

Each sticker carries TWO unique URLs that point to different endpoints in our web app. This is essential for our user experience and is a hard product requirement on the manufacturer.

Channel	URL pattern → Behaviour
QR Code (printed)	https://app.planum.de/p/{uuid} → public landing page; user registers / logs in, then sees the plant profile
NFC Chip (programmed)	https://app.planum.de/nfc/{signed_token} → secure auto-login: if the user is already logged in on the device, they jump straight into the plant profile; otherwise they receive a passwordless magic-link login

The manufacturer receives a CSV from us with three columns per sticker (uuid, qr_url, nfc_url). The QR press uses the qr_url column for variable printing; the NFC programmer uses the nfc_url column for chip encoding.

Backend security: the NFC token is signed and bound to a single sticker; it cannot be replayed by anyone who only takes a photo of the QR code. NFC chips must be locked against re-writing after programming.

4 • Weather Resistance — Hard Requirement

Because Planum covers outdoor and garden use from day one, every sticker (regardless of size) MUST be fully weatherproof for at least 3 years.

- Permanent water resistance — repeated watering, rain, condensation, full submersion ≥ 24 h
- UV stability — Blue Wool Scale > 6, no fading after 12 months direct sunlight
- Temperature range — -20 °C to +80 °C without adhesive failure or brittleness
- Soil & dirt — wipeable, no permanent staining
- Adhesion on terracotta, glazed ceramic, plastic, glass, powder-coated metal

Material preference: PET face material with matte lamination + UV protective overlamine. PVC acceptable as fallback if PET is not in the manufacturer’s portfolio. We are open to a counter-proposal from the supplier provided the durability targets are met.

5 • Technical Specifications

Parameter	Specification
Shape	Round, die-cut
Diameters	ø 25 mm / ø 30 mm / ø 40 mm — same SKU
Face material	PET ~80–100 µm (preferred), PVC alternative
Adhesive	Permanent acrylic, solvent-free, > 8 N/25 mm initial tack
Temperature	-20 °C to +80 °C operational
Printing	CMYK 4-colour, ≥ 600 dpi
Finishing	Matte lamination + UV protective overlamine

Water resistance	≥ 24 h immersion + permanent splash
Light fastness	Blue Wool Scale > 6
Durability	Min. 3 years indoor + outdoor
QR code	QR Model 2, ECC Level H (30 %), module ≥ 0.5 mm
QR variable data	Unique qr_url per sticker (CSV from Planum)
NFC chip	NXP NTAG 213 (180 byte) standard; NTAG 215 (504 byte) optional upgrade
NFC frequency / protocol	13.56 MHz, ISO/IEC 14443A
NFC programming	Pre-programmed unique nfc_url per chip, locked against re-writing
Certifications	REACH, RoHS, EU 10/2011, NFC chip CE / FCC

6 • Packaging — End-Customer Pack

Each end-customer pack contains 20 stickers in a fixed mixed-size assortment, plus printed materials:

- 8 × ø 25 mm (small pots, herbs, propagation)
- 8 × ø 30 mm (standard houseplants)
- 4 × ø 40 mm (garden / outdoor)
- Carrier sheet (A6 or folded A5), pre-arranged by size
- Branded paper sleeve / folded carton with Planum artwork (FSC paper preferred)
- 1 × instruction insert (DIN A6, 120 gsm, double-sided CMYK)
- Plastic-free outer wrap preferred

Shipping carton: 50 end-customer packs per outer carton, with lot number and production date printed on the outside.

Open packaging questions for the manufacturer (asked explicitly in the quotation request): can custom packaging be produced from artwork supplied by Planum, which file formats are accepted (we can deliver AI / PDF/X-1a / EPS / SVG), is there a library of ready-to-customise packaging templates we could choose from, and what colour profile (e.g. Fogra39) and bleed are required.

7 • Quantities & Pricing Targets

7.1 Pilot Order

Item	Quantity
Total stickers (one SKU)	1,000 pcs
End-customer packs	50 packs of 20 stickers
Size mix per pack	8 × ø25 mm + 8 × ø30 mm + 4 × ø40 mm
Size mix total	400 × ø25 mm + 400 × ø30 mm + 200 × ø40 mm

Retail price per pack	€29.00 (hardware) + €10.00/month optional Pro app add-on
Target landed cost / sticker	≤ €0.55 incl. packaging + shipping + duties

7.2 Follow-Up Orders (planned)

- 5,000 pcs — first re-order after market validation
- 10,000 pcs — Q3 scale-up
- 25,000 pcs — Q4 / holiday season
- 50,000 pcs — annual run-rate target

8 • Manufacturer Requirements

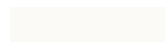
8.1 Mandatory

- Acceptance of CSV with 1,000+ unique entries (uuid, qr_url, nfc_url) for variable QR printing AND NFC programming
- NFC chips pre-programmed with unique nfc_url per UUID, locked against re-writing
- Fully assembled test pack (1–2 end-customer sets) before main production
- Digital proof / print preview before sign-off
- Test reports: water immersion, UV exposure, peel adhesion on terracotta
- Custom packaging from Planum artwork — sleeve, carrier, instruction insert
- Shipping options: DHL/FedEx/UPS air courier (DDP Germany) AND CIF Hamburg quoted separately

8.2 Preferred

- FSC-certified paper for sleeves and carrier; plastic-free outer wrap
- In-house warehousing / fulfilment option for follow-up orders
- Documented experience with combined QR + NFC stickers (reference projects)
- NDA on request

9 • Brand Colours

						
Forest #1F4D2C	Leaf #2D5F3F	Sage #87A878	Mist #C7D7C2	Cream #F5F1E8	Paper #FBFAF6	Ink #15241B

Final print-ready vector files (AI / PDF/X-1a) will be delivered after the manufacturer has confirmed accepted file formats and required colour profile.

10 • Shipping & Payment Terms

Sample / test pack	Within 14 days of order confirmation, via DHL Express
Main production lead time	Max. 30 days from sample approval
Pilot shipping (~2 kg)	Air courier preferred (DDP Germany incl. customs + 19 % VAT)

Follow-up shipping	Sea LCL CIF Hamburg + air courier alternatives, both quoted
Payment	Alibaba Trade Assurance preferred; alternative 30 % / 70 %
Warranty	Replacement or credit for defect rates > 2 %